

Problem

Obtain the transfer function of a high-pass filter with a passband gain of 100 and a cutoff frequency of 40 rad/s.

Step-by-step solution

Step 1 of 3

For a high pass filter having a pass band gain of 100 and cutoff frequency of $\omega_c = 40$ rad/s .

The expression for high pass filter pass band gain is,

$$K = -\frac{R_f}{R_i}$$

Substitute 100 for K in the above equation.

$$100 = \frac{R_f}{R_i}$$

$$R_f = 100R_i \dots\dots (1)$$

Step 2 of 3

The expression for a high pass filter cutoff frequency is,

$$\omega_c = \frac{1}{R_i C_i}$$

Substitute 40 rad/s for ω_c in the above equation.

$$40 = \frac{1}{R_i C_i}$$

$$R_i = \frac{1}{40C_i} \dots\dots (2)$$

Recall equation (1).

$$R_f = 100R_i$$

Substitute $\frac{1}{40C_i}$ for R_i in the above equation.

$$R_f = \frac{100}{40C_i}$$

$$R_f C_i = 2.5 \dots\dots (3)$$

Step 3 of 3

The transfer function of an active high pass filter is,

$$H(\omega) = \frac{-j\omega C_i R_f}{1 + j\omega C_i R_i}$$

Substitute 2.5 for $R_f C_i$ and $\frac{1}{40}$ for $R_i C_i$ in the above equation.

$$\begin{aligned} H(\omega) &= \frac{-j\omega(2.5)}{1 + j\omega\left(\frac{1}{40}\right)} \\ &= \frac{-j\omega(2.5)}{\frac{40 + j\omega}{40}} \\ &= \frac{-j\omega(2.5)(40)}{40 + j\omega} \\ &= \frac{-j\omega 100}{40 + j\omega} \end{aligned}$$

Therefore, the transfer function, $H(\omega)$ of the high pass filter is $\boxed{\frac{-j\omega 100}{40 + j\omega}}$.